The Dynamic Drone Routing Problem

Modelling, Objective and Trajectory-Based Metaheuristics

MetaLab Suite

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Abstract

This document develops the theoretical framework for the *Dynamic Drone Routing Problem* (DDRP), following the drone-delivery vehicle-routing model of Dorling et al. (2016). A fleet of unmanned aerial vehicles, launched from a depot, serves delivery orders that are revealed *in real time*. Each order carries a payload, and every drone is limited by a strict payload capacity Q and by a battery budget whose consumption scales with both the payload carried and the distance flown. We give a complete mixed-integer formulation of the static core, define the objective as the sum of travelled distance and unserved-order penalties, state the strict constraints, extend the model to the online setting through a rolling-horizon decomposition, and close with the design of tabu-search and iterated-local-search metaheuristics for the problem.

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1 Introduction

Unmanned aerial vehicles (drones) are increasingly used for last-mile parcel delivery. Dorling et al. [1] introduced and studied the *Vehicle Routing Problem for drone delivery*: a fleet of multi-rotor drones, launched from a depot, serves a set of customers under hard limits on payload weight and battery energy. Their analysis shows that the cost of a delivery is governed *jointly* by distance and by payload-dependent energy consumption, so routing and loading decisions are inseparable.

In the *Dynamic* Drone Routing Problem studied here, orders are not known in advance: they arrive over a planning horizon and must be served as they appear. A dispatcher must therefore

repeatedly reassign drones, trading off exploitation of the current plan against re-optimisation as new information arrives. The static problem generalises the classical vehicle-routing problem and is NP-hard [4]; the online nature makes it harder still [5].

2 The static core: sets, parameters and variables

Table 1: DDRP data and decision variables.

Symbol	Meaning
$\mathcal{K} = \{1, \dots, K\}$	set of drones
$\mathcal{I} = \{1, \dots, n\}$	set of delivery orders (customers)
0	the depot (*departure and return node*)
$V = \{0\} \cup \mathcal{I}$	node set
d_{ij}	travel distance between nodes i, j
w_i	payload (parcel weight) of order i
Q	payload capacity of a drone
B	usable battery energy of a drone
a	empty-drone energy consumption per unit distance
b	payload energy sensitivity per unit distance
v	cruise speed (distance per unit time)
τ	service (hover/landing) time per order
λ	penalty for an unserved order
$x_{ijk} \in \{0, 1\}$	1 if drone k travels directly from i to j
$y_{ik} \in \{0, 1\}$	1 if drone k serves order i
$q_{ijk} \geq 0$	payload carried by drone k on arc (i, j)
$t_{ik} \geq 0$	auxiliary ordering variable of node i in route k
$z_k \in \{0, 1\}$	1 if drone k is used

3 Payload-dependent battery model

A multi-rotor's power requirement increases with its total mass, hence with the payload it carries. Following the linearisation used by Dorling et al. [1], the *energy consumed per unit distance* by a drone carrying a payload q is

$$e(q) = a + b\,q, \qquad a \geq 0,\ b \geq 0, \tag{1}$$

so that the energy spent traversing arc (i, j) with payload q is $e(q)\,d_{ij}$. The constant a is the empty-drone consumption and $b\,q$ is the extra consumption caused by the carried mass. This dependency is what distinguishes DDRP from a pure distance-minimisation problem: a heavy parcel carried far consumes disproportionately more battery.

4 Objective function

Let $y_{ik} = \sum_{j \in V} x_{ijk}$ indicate whether drone k serves order i. The dispatcher trades off total travelled distance against the number of orders left unserved. The objective—*minimise distance and unserved penalties*—reads

$$\min\ Z(X) = \underbrace{\sum_{k \in \mathcal{K}} \sum_{i \in V} \sum_{j \in V} d_{ij}\, x_{ijk}}_{\text{total travelled distance}} + \lambda \underbrace{\sum_{i \in \mathcal{I}} \Big(1 - \sum_{k \in \mathcal{K}} y_{ik}\Big)}_{\text{unserved orders}} \tag{2}$$

where $\lambda \gg \max_{ij} d_{ij}$ is chosen large enough that serving an order is always preferable to abandoning it, yet finite so that genuinely unreachable orders (given the battery budget) are penalised rather than making the instance infeasible. If energy is to be priced explicitly, as in [1], the objective is augmented with $c_e \sum_k \sum_{i,j} (a + b\, q_{ijk})\, d_{ij}\, x_{ijk}$, where c_e is the cost per unit of energy; we keep the two-term form (2) as the canonical teaching objective.

5 Constraints

The following constraints are *strict*: no plan may violate them.

(C1) Unique assignment and unserved accounting. Each order is served by at most one drone,

$$\sum_{k \in \mathcal{K}} y_{ik} \;\le\; 1, \qquad i \in \mathcal{I}. \tag{3}$$

(C2) Flow conservation and route structure. For every served node the in-degree equals the out-degree, and every used drone leaves and re-enters the depot,

$$\sum_{j \in V} x_{ijk} = \sum_{j \in V} x_{jik} = y_{ik}, \quad i \in \mathcal{I},\ k \in \mathcal{K}; \qquad \sum_{j \in \mathcal{I}} x_{0jk} = \sum_{i \in \mathcal{I}} x_{i0k} = z_k, \quad k \in \mathcal{K}. \tag{4}$$

(C3) Payload capacity. The payload flowing into a customer equals the payload flowing out plus the parcel delivered there; the depot loads each drone with exactly the parcels it will deliver:

$$\sum_{j \in V} q_{ijk} - \sum_{j \in V} q_{jik} = -\, w_i\, y_{ik}, \qquad i \in \mathcal{I},\ k \in \mathcal{K}, \tag{5}$$

$$\sum_{j \in V} q_{0jk} = \sum_{i \in \mathcal{I}} w_i\, y_{ik}, \qquad k \in \mathcal{K}, \tag{6}$$

and the capacity limit holds on every arc and for the whole load,

$$0 \;\le\; q_{ijk} \;\le\; Q\, x_{ijk}, \quad i, j \in V,\ k \in \mathcal{K}; \qquad \sum_{i \in \mathcal{I}} w_i\, y_{ik} \;\le\; Q\, z_k, \quad k \in \mathcal{K}. \tag{7}$$

(C4) Battery budget. The energy consumed along the whole route, with consumption scaling with payload (1) and distance, must not exceed the usable battery energy,

$$\sum_{i \in V} \sum_{j \in V} (a + b\, q_{ijk})\, d_{ij}\, x_{ijk} \;\le\; B\, z_k, \qquad k \in \mathcal{K}. \tag{8}$$

In practice one reserves a safety margin B_{res} for wind, hovering and landing, replacing B by $B - B_{\text{res}}$.

(C5) Subtour elimination. Auxiliary ordering variables with a big-M bound forbid disconnected subtours,

$$t_{ik} - t_{jk} + n\, x_{ijk} \;\le\; n - 1, \qquad i, j \in \mathcal{I},\ i \neq j,\ k \in \mathcal{K}. \tag{9}$$

(C6) Release times (dynamic information). An order i with release time r_i may be served only once it is known. With t^a_{ik} the arrival time at order i,

$$y_{ik} = 1 \;\Rightarrow\; t^a_{ik} \;\ge\; r_i, \tag{10}$$

linearised as $t^a_{ik} \ge r_i - M(1 - y_{ik})$. Optional time windows $[\underline{e}_i, \overline{e}_i]$ extend this to $\underline{e}_i \le t^a_{ik} \le \overline{e}_i$ when $y_{ik} = 1$.

Remark 1. *Constraints (C1)–(C5) define the static DDRP. Its feasible region is discrete and its objective* (2) *is a linear function of the decision variables, so the static core is a (NP-hard) mixed-integer program.*

6 The dynamic setting

In DDRP the order set $\mathcal{I}$ is not known a priori. Orders arrive according to a stochastic process—typically a non-homogeneous Poisson process with rate $\nu(t)$ over a horizon $[0, T]$—and each order becomes *actionable* at its release time r_i. The dispatcher observes the state

$$S_t = (\text{drone positions},\ \text{remaining batteries},\ \text{served orders},\ \text{pending orders}) \tag{11}$$

and selects an action: dispatch a drone, insert a customer into an existing route, or idle. A *policy* π maps states to actions, and the Dynamic DDRP seeks

$$\min_{\pi}\ \mathbb{E}\Big[\ \sum_{t=0}^{T}\Big(\textstyle\sum_k \sum_{i,j} d_{ij}\, x_{ijk}^{t}\ +\ \lambda\, U_t\Big)\ \Big], \tag{12}$$

where U_t is the number of still-unserved, already-released orders at epoch t, and the expectation is over the unknown arrival process.

A tractable and widely used approach is *rolling-horizon re-optimisation*: at each decision epoch the dispatcher solves a *static* instance consisting of all pending orders plus new arrivals, using the drones' current positions, remaining payload and remaining battery as initial conditions, and commits to the first portion of the plan before re-optimising at the next epoch. This turns the online problem into a sequence of offline DDRP instances, each solvable by the metaheuristics below.

7 Trajectory-based metaheuristics for the DDRP

Encoding. A convenient single-solution encoding is the *giant tour*: a permutation of the orders, split (optimally, by dynamic programming) into K drone routes. The split respects capacity (7) and battery (8) and is recomputed after every move. Orders the split cannot place are unserved and are charged λ in (2).

Neighbourhoods. Classic routing moves operate on the permutation:

- **relocate** — remove one order and reinsert it elsewhere;
- **exchange** — swap two orders;
- **Or-opt** — move a chain of two or three consecutive orders;
- **2-opt** — reverse a segment within a route.

Algorithm 1 Tabu search for the (static) DDRP

$s \leftarrow$ CheapestInsertion(pending orders)
$s^\star \leftarrow s$; $\mathcal{T} \leftarrow \emptyset$
while stopping criterion not met **do**
  $s' \leftarrow \arg\min_{\hat{s} \in \mathcal{N}(s)} \{Z(\hat{s}) : \text{move}(\hat{s}) \notin \mathcal{T}\}$
  if s' is tabu but $Z(s') < Z(s^\star)$ **then** ▷ aspiration
    release s' from $\mathcal{T}$
  end if
  $s \leftarrow s'$; insert the reverse move into $\mathcal{T}$ with tenure θ
  if $Z(s) < Z(s^\star)$ **then**
    $s^\star \leftarrow s$
  end if
end while
return $s^\star$

Why trajectory methods suit the dynamic setting. Trajectory methods are the natural choice for DDRP because (i) they are *anytime*, so the dispatcher can halt them when a new order arrives and use the best solution available; (ii) they *warm-start* trivially, by seeding the search with the previous plan so that real-time re-optimisation inherits past effort; and (iii) the giant-tour representation admits cheap, incremental evaluation of relocate and exchange moves. A common practical recipe is tabu search [2] with the moves above, occasionally perturbed à la iterated local search [3] to diversify.

8 Complexity and closing remarks

The static DDRP contains the capacitated vehicle-routing problem as a special case and is therefore NP-hard [4]; the payload-dependent energy model (1) strengthens the coupling between routing and loading. The dynamic version is an online stochastic optimisation problem whose myopic solutions are provably suboptimal in general [5]; good policies combine rolling-horizon re-optimisation with a fast, anytime trajectory metaheuristic. Adaptive large-neighbourhood search [7] and scenario-based stochastic planning [6] provide strong baselines for comparison.

References

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